



MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : BS-BIO301 Biology

UPID : 003488

Time Allotted : 3 Hours

Full Marks : 70

The Figures in the margin indicate full marks.

Candidate are required to give their answers in their own words as far as practicable

Group-A (Very Short Answer Type Question)

1. Answer any ten of the following :

[1 x 10 = 10]

- ☒ (i) Who is called the "Father of Taxonomy"?
- ☒ (ii) _____ are colourless plastids and mainly help in storage of food compounds.
- ☒ (iii) The chromosomes with satellite are known as _____.
- ☒ (iv) A codon contain how many nucleotides?
- ☒ (v) The nature of an enzyme is _____.
- ☒ (vi) Mitosis occurs in our _____ cells.
- ☒ (vii) Rough Endoplasmic Reticulum contains _____.
- ☒ (viii) H_2 donor during photosynthesis is _____.
- ☒ (ix) Lacto bacillus bacteria are responsible for curd formation.
- ☒ (x) Brownian motion discovered by _____.
- ☒ (xi) In case of _____ organisms, a single cell carries out all necessary life processes.
- ☒ (xii) Modern humans have 46 chromosomes. Their sperms and eggs will have 23 chromosomes each.

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

- ☒ 1. How an enzyme works to speed up a reaction? [5]
- ☒ 2. Explain the structure of nucleosome with diagram. [5]
- ☒ 3. Draw labeled diagram of Animal cell as seen in Electron microscope. Comment on characteristics of Animal cell. [5]
- ☒ 4. State five variations on Mendel's rules. [5]
- ☒ 5. Explain the function of Intracellular receptors with diagram. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

- ☒ 6. What are Model organisms? Give characteristics of *C. elegans*, *D. Melanogaster* and *M. musculus* as model organisms. fruit fly mouse [15]
- ☒ 7. Write a short note on 'Sex Determination of Human'. What are Multiple alleles and Lethal alleles? [10+5]
- ☒ 8. Write the name and functions of main three different Types of RNA. What are phospholipid and glycolipids? Differentiate between saturated and unsaturated fats. [6+4+5]
- ☒ 9. Define enzyme inhibition. Explain the different types of reversible inhibitions with suitable diagram. Explain the factors affecting enzyme activity. [2+8+5]
- ☒ 10. What is DNA replication? State in brief the mechanism of DNA replication in prokaryotes with suitable diagram. [2+13]

*** END OF PAPER ***